

Performance, Testing & Quality Gates

Optimization plan that protects visual fidelity and prevents demo-day failures

1. Performance targets

- Fast first paint on a normal laptop/mobile connection; optimize the landing hero and font loading first.
- Zero avoidable cumulative layout shift from images, fonts or validation messages.
- Typing should remain instant; form validation must not trigger expensive whole-page rerenders.
- Keep the assessment bundle lean: remove unused UI libraries and duplicate icon packs.
- Use WebP/AVIF for large raster assets where practical and explicit width/height or aspect-ratio to reserve space.

2. Quality test pyramid

Layer	What to test	Suggested tool
Unit	normalizers, schemas, age rules, dependent-option filtering	Vitest
Component	field errors, loading button, OTP behavior, progress	React Testing Library
Flow	happy path, invalid path, back navigation, failure/retry	Playwright
Manual visual	reference parity at key breakpoints	Side-by-side screenshots
Accessibility	keyboard, labels, contrast, zoom, reduced motion	axe + manual keyboard pass

3. Mandatory flow scenarios

1. Fresh landing → terms → return/continue → complete all four signup steps successfully.
2. Invalid email → correction → verification loading → success.
3. Verification failure → retry without losing email.
4. OTP incomplete/wrong → error → correct OTP → continue.
5. Back from step 3 to step 2 → edit value → forward again with state preserved.
6. Under-age or invalid age behavior.
7. Dependent state/city/college change behavior if those fields exist in the reference.
8. Final submit failure → retry → success.
9. Mobile keyboard and autofill behavior.
10. Desktop and tablet resizing without broken composition.

4. Visual regression checklist

- Compare reference and web screenshots at the same logical screen/state.
- Check typography: family, weight, size, line-height and letter spacing.
- Check spacing: page gutters, card padding, field gaps, CTA spacing.
- Check component geometry: heights, radii, borders, shadows and icon alignment.
- Check background crop/position at each breakpoint.
- Check disabled/loading/error states, not only the happy path.

5. Lighthouse-oriented optimization

- Preconnect only when a real external origin is necessary; self-host/localize assets when licensing permits.
- Use font-display: swap and limit Poppins weights to those actually used.
- Do not ship giant animation/video libraries for a signup wizard.
- Use route-level code splitting only if it measurably helps.
- Provide alt text for meaningful imagery and empty alt for decorative images.
- Ensure buttons/links have accessible names so accessibility audits remain clean.

6. Release gate

Do not record the submission until: production build succeeds; no console errors; every validation branch is reproducible; mobile/tablet/desktop pass; keyboard flow passes; the simulated failure trigger is known; and the entire demonstration fits comfortably inside five minutes.

5-minute recording plan

Time	Show
0:00–0:35	Landing + responsive quality + Terms entry.
0:35–1:10	Terms + navigation into signup.
1:10–2:10	Step 1 invalid email, correction, loading, verification/OTP.
2:10–3:35	Steps 2–4, one edge case, Back navigation and preserved state.
3:35–4:20	Simulated submission failure + retry + success.
4:20–4:50	Quick mobile/desktop responsive resize and polished states.
4:50–5:00	End on success screen; no source-code tour unless requested.